Project Title: EV Charging Energy Consumption Prediction

GMR Institute of Technology (GMRIT)

Team Name: Team Ignis

Team Details

#	Name	Roll Number	Role
1	L. Pranay	23341A1266	Team Lead
2	P. Srikar	23341A1288	Member
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Repository Links

- L. Pranay – https://github.com/pranaylekkala5390/GenAI-Assignment
- P. Srikar – https://github.com/srikar0119/GenAI-Assignment
- S. Haneesh – https://github.com/Haneesh-Srirama/GenAI-Assignments
- V. Bhargav Raju – https://github.com/bhargav123321/GenAI-Assignments

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Project Updates

1. Data Inspection

Dataset Overview

Number of rows: 8,354

Number of columns: 27

Dataset dimensions: 8,354 × 27

Duplicate rows: 0

Missing values: 0 (every column fully populated)

Feature Groups

- 12 numerical features — battery capacity, initial state of charge (SOC), charging power, queue length, station load, electricity price, and more
- 9 categorical features — vehicle type, station ID, location type, weather, traffic density, day of week, and more
- Timestamp columns retained for downstream feature engineering

Target Variable

energy_consumed_kWh — the amount of energy (in kWh) consumed during a charging session, to be predicted using only information available before or at the start of charging.

2. Data Preprocessing

Handling Missing Values

Missing values were checked across all 8,354 sessions.

Observation: 0 missing values were found in the dataset.

Duplicate Records

Duplicate records were checked across the dataset.

Observation: 0 duplicate rows were found.

Leakage Removal

The following columns were dropped prior to modeling because they are algebraically derivable from the target variable and would leak information about the outcome:

- final_soc
- charging_duration

Feature Engineering

New time-based features were extracted from the arrival_time timestamp:

- arrival_hour
- day_of_week
- is_weekend
- time_period

Outlier Decision

High-energy charging sessions were reviewed and retained as valid data points, since they correspond to large-battery vehicles (e.g., buses) rather than data-entry errors.

Final Dataset

Rows: 8,354

Columns: 27

Missing values: 0

Duplicate records: 0

3. Data Visualization

A total of 7 charts were built using Matplotlib and Seaborn (3 chart types: Histogram, Box Plot, Bar Chart) to explore how battery, charging, and station-level factors relate to energy consumption, and to evaluate the final model's predictions.

Energy Consumption Distribution

Chart: Histogram

Purpose: To understand the frequency of sessions across different energy-consumption levels

Observation: Right-skewed distribution — most sessions fall in the 20–55 kWh band (mean 39.1 kWh, median 35.1 kWh)

Energy Consumed by Battery-Capacity Group

Chart: Bar Chart

Purpose: To compare average energy consumed across battery-capacity groups (30, 40, 60, 75 and 100 kWh packs)

Observation: Larger battery-capacity groups consume more energy per session — the strongest predictor of energy consumption

Energy Consumed by Initial SOC Group

Chart: Bar Chart

Purpose: To compare average energy consumed across initial state-of-charge groups (10–60%, in 10% bands)

Observation: Lower initial-SOC groups consume more energy — an inverse relationship consistent with charging logic

Energy Consumed by Vehicle Type

Chart: Box Plot

Purpose: To visualize the spread of energy consumption across Bus, Car, and Two-Wheeler sessions

Observation: Similar median consumption (~35 kWh) across vehicle types; retained as a categorical feature

Energy Consumed by Charging-Power Group

Chart: Bar Chart

Purpose: To compare average energy consumed across charging-power groups (7, 11, 22 and 50 kW)

Observation: A weak, gradual relationship — power alone explains only part of the variation but is included since it is known pre-session

Model Performance Comparison

Chart: Bar Chart

Purpose: To compare MAE and R^2 across the three trained models — Linear Regression, Random Forest, and Gradient Boosting

Observation: Gradient Boosting wins on both metrics (MAE 0.97 kWh, R^2 0.994) and was selected as the final model

Feature Importance

Chart: Bar Chart

Purpose: To compare the relative importance of each input feature in the final model

Observation: Battery capacity and initial SOC dominate, guiding which inputs matter most in deployment

4. Model Training

Train/Test Split

An 80% train / 20% test split was used, with a fixed random_state for reproducible and comparable results.

Shared Preprocessing Pipeline

An identical encoding and scaling pipeline was applied to all three models before fitting, ensuring the comparison between models stays fair.

Models Trained

- Linear Regression (baseline)
- Random Forest
- Gradient Boosting

Target Variable

energy_consumed_kWh, predicted using only features available before or at the start of charging.

5. Evaluation & Selection

Metrics Used

Mean Absolute Error (MAE) and R^2 score, both computed on the held-out test set.

Model Comparison

Model	MAE (kWh)	R^2 Score
Linear Regression	Higher	Lower
Random Forest	Moderate	Moderate

Model	MAE (kWh)	R^2 Score
Gradient Boosting	0.97	0.994

Final Model Selected

Model: Gradient Boosting

Performance: MAE 0.97 kWh, R^2 0.994 on unseen test sessions

Why it wins: Captures the non-linear relationships between battery capacity, SOC, and power that the linear model misses

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Project Workflow Summary

- Step 1 — Data Inspection: 8,354 rows × 27 columns reviewed; 0 missing or duplicate values
- Step 2 — Cleaning & Leakage Removal: deterministic leakage columns dropped before modeling
- Step 3 — Feature Engineering: time-based features extracted from the arrival timestamp
- Step 4 — Exploratory Analysis: 7 charts built to study the drivers of energy consumption
- Step 5 — Model Training: 3 regression models trained through one shared pipeline
- Step 6 — Evaluation & Selection: Gradient Boosting selected — MAE 0.97 kWh, R^2 0.994

Tools & Libraries Used

- Python
- Pandas
- Scikit-learn
- Matplotlib
- Seaborn